

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 07

Miscellaneous Topics



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Recap of Previous Lecture



Functionally complete function

A collection of various tools and cables arranged in a circular pattern around a central dark blue circle. The tools include pliers, wire cutters, a screwdriver, and a pair of scissors. The cables are of various colors (red, blue, green, yellow, black) and are bundled together.

Topics to be Covered

Question Discussion



[NAT]

$$2^{-1} 2^{-2} 2^{-3} 2^{-4}$$

If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of $x + y$ is 3. [GATE-2021-CS: 1M]

$$= \frac{1}{2} \times 1 + \frac{1}{4} \times 1 + 0 + \frac{1}{16}$$

$$= 0.5 + 0.25 + 0.0625$$

$$= (0.8125)_{10}$$

$$x=1$$

$$y=2$$

[NAT]



Consider a quadratic equation $x^2 - \underline{13}x + 36 = 0$ with coefficients in a base b . The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b = \underline{8}$.

[GATE-2017-CS: 1M]

$x^2 - \text{sum of roots } x + \text{multiplication of roots} = 0$

$$x = (5)_b, x = (6)_b$$

$$\begin{array}{r} (5)_8 \\ + (6)_8 \\ \hline (13)_8 \end{array}$$

$$x = (5)_b = (5)_{10}$$

$$(13)_b = (b+3)_{10}$$

$$(36)_b = (3b+6)_{10}$$

$$[x^2 - (b+3)x + (3b+6)]_{10} = 0$$

$$x = (5)_{10}$$

$$25 - (b+3)5 + 3b+6 = 0$$

$$-2b = -16$$

$$b = 8$$

$$[x^2 - 11x + 30]_{10} = 0$$

$$(x-6)(x-5) = 0 \rightarrow x = (6)_{10}, (5)_{10}$$



[NAT]

Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solution is 5. [GATE-2015-CS: 2M]

$$\begin{matrix} x^1 & x^0 \\ (4 & 3) \end{matrix}_x = \begin{matrix} 8^1 & 8^0 \\ (y & 3) \end{matrix}_8$$

$$x \geq 5 \quad \& \quad 0 \leq y \leq 7$$

$$(4x+3)_{10} = (8y+3)_{10}$$

$$x = 2y$$

$$3 \leq y \leq 7$$

$$x = (6, 8, 10, 12, 14)$$

[NAT]



The base (or radix) of the number system such that the following equation holds is 5. $312/20 = 13.1$

[GATE-2014-CS: 1M]

$$\hookrightarrow b \geq 4$$

$$\frac{312}{20} = 13.1$$

$\begin{matrix} b^2 & b^1 & b^0 \\ (312)_b = (3b^2 + b + 2)_{10} \\ (20)_b = (2b)_{10} \\ (13.1)_b = (b + 3 + 1/b) \end{matrix}$

$$\frac{3b^2 + b + 2}{2b} = b + 3 + 1/b$$
$$3b^2 + b + 2 = 2b(b + 3 + 1/b)$$
$$3b^2 + b + 2 = 2b^2 + 6b + 2$$
$$b^2 - 5b = 0$$
$$b(b - 5) = 0$$
$$b = 0, 5 \checkmark$$

X

[NAT]



Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is 3.

[GATE-2014-CS: 1M]

$$y \geq 9 \quad \& \quad y \geq (x+1)$$

$$y \geq 9 \quad \& \quad x \leq (y-1)$$

$$(123)_5 = \overset{y^1 y^0}{(x \ 8)}_y$$

$$(25 + 10 + 3)_{10} = (xy + 8)_{10}$$

$$xy = 30$$

$$y=10 \longrightarrow x=3 \checkmark$$

$$y=15 \longrightarrow x=2 \checkmark$$

$$y=30 \longrightarrow x=1 \checkmark$$

[MCQ]



Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r} = 11_r$ is $\longrightarrow r \geq 3 \longrightarrow r > 2$ [GATE-2008-CS: 1M]

- ☐ A Decimal 10
- ☐ B Decimal 11
- ☐ C Decimal 10 and 11
- ☒ D Any value > 2

$$\sqrt{(r^2 + 2r + 1)_{10}} = (r + 1)_{10}$$

$$r^2 + 2r + 1 = (r + 1)^2 \longrightarrow \text{identity} \longrightarrow \text{satisfies for all the values of } r$$

✓✓ Overall $r \geq 3$

[MCQ]



Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A=01010$ and $B = 11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow? [GATE-2024-CS: 1M]

☒ A $A + B \Rightarrow (4)_{10}$

☒ B $A - B (16)_{10}$

☐ C $B - A (-16)_{10}$

☐ D $2 * B (-12)_{10} \quad (B+B)$

$-(16)_{10} \text{ to } (15)_{10}$

$A = 01010 = (+10)_{10}$

$B = 11010 = (-6)_{10}$

[MCQ]



In 16-bit 2's complement representation, the decimal number -28 is:

[GATE-2019-CS: 1M]

- ☐ A 1111 1111 0001 1100 ✗
- ☐ B 0000 0000 1110 0100 ✗
- ☒ C 1111 1111 1110 0100 ✓
- ☐ D 1000 0000 1110 0100 ✗

$$\begin{aligned} & \text{6 bit} \\ \Rightarrow +28 &= (011100)_2 \\ & \downarrow 2's \\ & (100100)_2 \\ &= (1111111111111111100100)_2 \\ (-28) &= (100100)_2 \end{aligned}$$

[NAT]



Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

[GATE-2019-CS: 1M]

[MCQ]

Consider $Z = X - Y$, where X , Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

[GATE-2019-CS: 1M]

- ☐ A n bits
- ☐ B $n-1$ bits
- ☒ C $n+1$ bits
- ☐ D $n+2$ bits

[MCQ]



Consider the unsigned 8-bit fixed point binary number representation below,

$$\boxed{b_7 \ b_6 \ b_5 \ b_4 \ b_3} \cdot \overset{2^{-1}}{b_2} \overset{2^{-2}}{b_1} \overset{2^{-3}}{b_0}$$

where the position of the binary point is between b_3 and b_2 . Assume b_7 is the most significant bit. Some of the decimal numbers listed below cannot be represented exactly in the above representation:

(i) 31.500 ✓ (ii) 0.875 ✓ (iii) 12.100 ✗ (iv) 3.001 ✗✗

Which one of the following statements is true?

[GATE-2018-CS: 1M]

- ☒ A None of (i), (ii), (iii), (iv) can be exactly represented
- ☒ B Only (ii) cannot be exactly represented
- ☒ C Only (iii) and (iv) cannot be exactly represented
- ☒ D Only (i) and (ii) cannot be exactly represented

$$= \frac{1}{2} b_2 + \frac{1}{4} b_1 + \frac{1}{8} b_0$$

$\swarrow$	$\searrow$	$\swarrow$	$\searrow$	$\swarrow$	$\searrow$
0.5	0	0.25	0	0.125	0
<u>0.5</u>		<u>0.25</u>		<u>0.125</u>	

[MCQ]



When two 8-bit numbers $A_7 \dots A_0$ and $B_7 \dots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a ripple-carry adder, the sum bits obtained are $S_7 \dots S_0$ and the carry bits are $C_7 \dots C_0$. An overflow is said to have occurred if

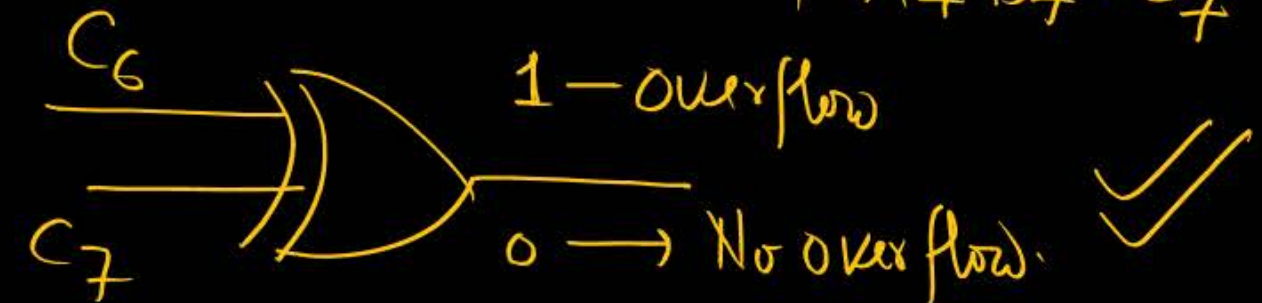
[GATE-2017-CS: 1M]

- ☒ A the carry bit C_7 is 1
- ☒ B all the carry bits ($C_7, \dots, C_0$) are 1
- ☒ C $(A_7 \cdot B_7 \cdot \bar{S}_7 + \bar{A}_7 \cdot \bar{B}_7 \cdot S_7)$ is 1.
- ☒ D $(A_0 \cdot B_0 \cdot \bar{S}_0 + \bar{A}_0 \cdot \bar{B}_0 \cdot S_0)$ is 1.

A_7
 B_7
 S_7

Result

$$\text{Overflow} = \bar{A}_7 \bar{B}_7 \cdot S_7 + A_7 B_7 \cdot \bar{S}_7$$



[NAT]



Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation. Then $X-Y$ is 1.

[GATE-2016-CS: 1M]

$$X = 2^{16} \quad - (2^{15}) \quad \text{to} \quad (2^{15}-1)$$

$$Y = (2^{16}-1) \quad - (2^{15}-1) \quad \text{to} \quad (2^{15}-1)$$

$$X-Y = 1$$



[NAT]

The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is -11. [GATE-2016-CS: 1M]

$$= (10101)_2$$

$$= (-16 + 5)$$

$$= (-11)_{10}$$

[MCQ]

The smallest integer that can be represented by an 8-bit number in 2's complement form is

[GATE-2013-CS: 1M]

- ☐ A -256
- ☒ B -128
- ☐ C -127
- ☐ D 0

$$\begin{aligned}
 & - (2^7) \text{ to } (2^7 - 1) \\
 & = (-128 \text{ to } 127)
 \end{aligned}$$

[MCQ]

P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$.

The 2's complement representation of $8 \times P$ is

[GATE-2010-CS: 1M]

- ☒ A $(C3D8)_{16}$
- ☐ B $(187B)_{16}$
- ☐ C $(F878)_{16}$
- ☐ D $(987B)_{16}$

$$\begin{aligned}
 P &= (1111\ 1000\ 0111\ 1011)_2 \\
 8 \times P &= (\underbrace{1111}_{\text{shift left 3}}\ 1000\ 0111\ 1011\ 000)_2 \\
 &= (1100001111011000)_2 \\
 &= (C3D8)_{16}
 \end{aligned}$$

[MSQ]



Which of the following is/are EQUAL to 224 in radix-5 (i.e., base-5) notation?

[GATE-2014-CS: 1M]

- A 64 in radix-10 ✓✓
- B 100 in radix-8 ✓✓
- C 121 in radix-7 ✓✓
- D ✗ 50 in radix-16 = $(80)_{10}$

$$(224)_5 = 5^0 + 10 + 4 = (64)_{10}$$

(A, B, C)

$$= (100)_8 = (100)_8$$

$$= (121)_7$$

$$= (121)_7$$

[NAT]

The total number of prime implicants of the function $f(w,x,y,z) = \Sigma(0, 2, 4, 5, 6, 10)$ is 3.

[GATE-2022-CS: 1M]

	$\bar{y}\bar{z}$	$\bar{y}z$	$y\bar{z}$	yz
$\bar{w}\bar{x}$	1			1
$\bar{w}x$	1	1		1
$w\bar{x}$				
wx				1

$\Rightarrow$

PI $\rightarrow 3$

EPI $\rightarrow 3$

[NAT]



Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is 3. [GATE-2018-CS: 2M]

PI $\rightarrow 4$

EPI $\rightarrow 3$

	$\overline{R}\overline{S}$	$\overline{R}S$	RS	$R\overline{S}$
$\overline{P}\overline{Q}$	1		X	1
$\overline{P}Q$		1	1	
$P\overline{Q}$	X			X
PQ	X	1	1	X

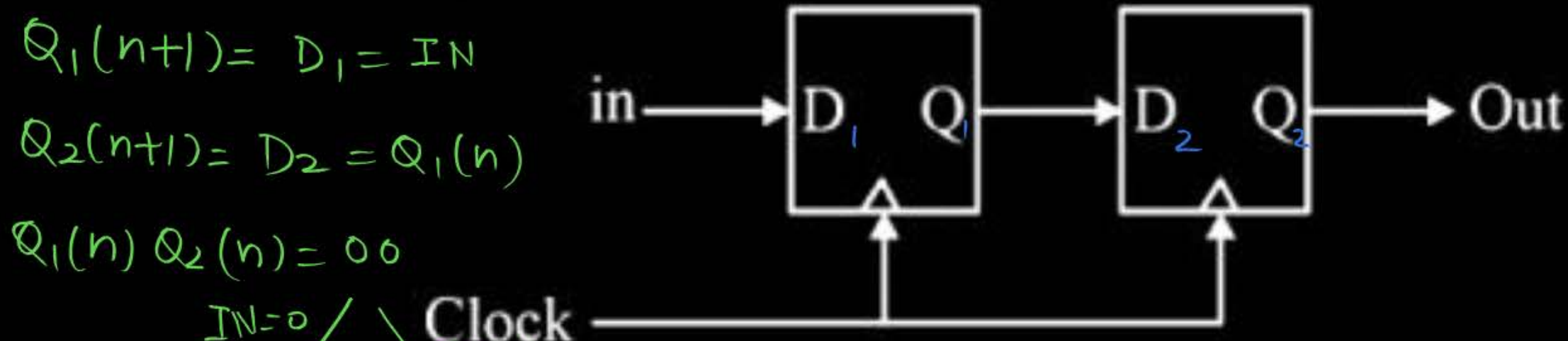
[NAT]



Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.

The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is 2.

[GATE-2018-CS: 1M]



$$Q_1(n+1) = D_1 = \text{IN}$$

$$Q_2(n+1) = D_2 = Q_1(n)$$

$$Q_1(n) Q_2(n) = 00$$

$$\begin{array}{l} \text{IN}=0 \swarrow \searrow \text{IN}=1 \\ 00 \quad 10 \end{array}$$

$$Q_1(n) Q_2(n) = 01$$

$$\begin{array}{l} \text{IN}=0 \swarrow \searrow \text{IN}=1 \\ 00 \quad 10 \end{array}$$

$$Q_1(n) Q_2(n) = 10$$

$$\begin{array}{l} \text{IN}=0 \swarrow \searrow \text{IN}=1 \\ 01 \quad 11 \end{array}$$

$$Q_1(n) Q_2(n) = 11$$

$$\begin{array}{l} \text{IN}=0 \swarrow \searrow \text{IN}=1 \\ 01 \quad 11 \end{array}$$

Thank you

GW
Soldiers!

